

The impact of congestion charging on car ownership: Evidence from a quasi-natural experiment

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ABSTRACT

Urban Vehicle Access Regulations are experiencing a resurgence in interest as practitioners grapple with prominent challenges, such as traffic safety, poor air quality, and inadequate liveability in cities. Congestion charging is one variant of such regulations, initially designed to minimise recurrent congestion formation and also holding the potential to produce ancillary impacts. The ability of congestion charging to discourage the ownership of private cars is an ancillary impact that has comparatively received limited research attention despite a direct relation to congestion. This study focusses on this gap by applying a quasi-natural experiment using the London Congestion Charge as a case study. Spatial variants of difference-in-difference models are applied to identify the causal impact of the congestion charge policy on private car registrations. In treatment group locations outside of the congestion charge boundary, the analysis indicates that private car registrations reduced by 6.95% following the introduction of the policy. The findings of this analysis could be helpful to policymakers considering the introduction of Urban Vehicle Access Regulations by providing quantitative evidence of some of the broader impacts such policies generate.

1. Introduction

Once, cars were seen as the embodiment of modernity. A mutually supportive complex of institutional actors, social trends, and industrial innovations cemented automobility as the dominant system of personal transit (Beckmann, 2001). In the mid-twentieth century, cities on both sides of the Atlantic demolished large swathes of residential areas to make space for highways (Safdie, 1998). Since the turn of the millennium, views have begun to change and there has been an increased application of policies designed to deemphasise the presence of cars within cities (Kuss and Nicholas, 2022). This trend is visible in initiatives that establish pedestrian zones in urban cores, introduce one-way streets in urban areas to reduce car throughput, remove street parking permits, and auction licenses to own cars within a city. Proponents of such strategies point to the discontents of automobility, such as the injuries and deaths caused by traffic collisions, the years of life lost to health conditions linked to air pollutants emitted by motorised traffic, and the severance of communities introduced by road infrastructures (Gössling, 2020).

Congestion charging is another such initiative designed to reduce frequent traffic movements within urban settings. Policies of this type

typically operate on a zone or cordon basis, requiring car travellers to pay a fee to cross the boundary (Ricci et al., 2017). Cities of considerably different situational contexts have introduced congestion charging, ranging from the small historic city of Durham in North-East England to the capital of the United Kingdom, London. The introduction of congestion charging provides scholars with an opportunity to evaluate different aspects of such transport policies, such as considerations of public acceptability, the effectiveness of the policy in reducing traffic volumes, as well as ancillary impacts that the policy may generate. For these ancillary impacts, attention has been paid to specific outcomes, such as air quality levels, retail sales, residential property values, road safety, and social equity (Beevers and Carslaw, 2005; Tang, 2021; Quddus et al., 2007; Noland et al., 2008; Bonsall and Kelly, 2005; Zheng et al., 2023).

One ancillary impact that has yet to be considered in detail is whether the introduction of congestion charging discourages car ownership, with only the work of Gonzalez et al. (2021) touching on this specific issue. As mentioned by Eliasson (2009), it is entirely plausible that changes in car ownership could occur as a longer-term impact of the introduction of congestion charging by alternating the relative cost implications of car trips as compared to alternative modes. This paper

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focusses on this particular issue by applying a quasi-natural experiment to the London Congestion Charge. Data on car ownership levels in areas that immediately surround the London Congestion Charge boundary are evaluated in a difference-in-differences analysis to determine whether introducing the policy reduced private car registrations. The contribution of this study is twofold. First, this study applies spatial variants of difference-in-differences to control for the occurrence of spatial autocorrelation to produce robust estimates of the policy impact. The study then compares the results of the spatial difference-in-differences experiment to non-spatial counterparts to showcase the benefits of accounting for spatial interaction when utilising geographically located outcome data. Second, the findings of the analysis provide supporting evidence to policymakers who desire to implement congestion charging in new areas (or adapt the operation of existing schemes) by showcasing the wider impacts that can arise from the initiative. This policy is still garnering considerable attention with transport planners, such as the new congestion charge being designed for Manhattan, New York (NY Gov, 2023). As cities in the rapidly expanding global south grapple with increasing rates of car ownership, it is likely that congestion charging policies will also be put on the table in these locations in the coming decade.

2. Background

This article connects with two vibrant areas of academic research: the application of congestion charging and the occurrence of car ownership. The following subsections summarise some popular themes in these areas to provide an access point for interested readers and position this study in the broader literature.

2.1. Congestion charging

Policies designed to modify the mobility choices and behaviours of citizens are generally referred to as transport demand management (TDM) initiatives. Such initiatives can be further classified based on whether their objective is to pull citizens towards desirable actions or push them away from detrimental practices (Gärling and Schuitema, 2007). Congestion charging is primarily a form of push-TDM, with drivers typically being discouraged from using their cars within certain areas by applying charges (or other forms of access restriction) to minimise excessive traffic movements. Congestion charging can also include elements of pull-TDM, for instance if the revenues generated are hypothecated into public and active travel improvements (Kottenhoff and Freij, 2009; Schuitema and Steg, 2008).

During the early phase of congestion charge development, academic research provided key insights on the planning and operational parameters of the policy. Whether the congestion charge would be economically viable to operate was a prominent feature, with applications of cost-benefit analysis being popular (Eliasson et al., 2009; Leape, 2005). In part, this issue of economic feasibility depended on the spatial extent of the policy. Determining the optimal boundary for a congestion charge to maximise the intended outcomes of the policy assisted in deciding on these design questions (Sumalee et al., 2009). As congestion charges started to move off drawing-boards towards real-world introductions, considerations of public reaction to the schemes took centre stage (Nikitas et al., 2018). On this issue, applications of psychometric models were prominent as researchers aimed to identify the antecedents which feed into the formation of public acceptability (Golob, 2003). A mixture of traditional psychological variables was found to hold sway, such as personal norms, alongside some novel antecedents, such as perceptions of policy fairness and efficacy (Eriksson et al., 2006, 2008; Kim et al., 2014). Outside of psychological variables, car ownership itself has also been found to be a primary factor in public acceptability of congestion charging, with car owners significantly less likely to back the introduction of such policies (Li et al., 2020).

After implementation of the policy, research attention began to shift

towards evaluating the observable outcomes (Axsen and Wolinetz, 2021). Following the intended objective of congestion charging, assessments of traffic flow impacts were prominent. Here, the literature paints a reasonably consistent picture of traffic volumes reducing post-introduction with subsequent improvements in journey time reliability and delays (Börjesson et al., 2012; Gibson and Carnovale, 2015; Santos, 2005). However, the initial reductions in congestion post-policy implementation may not be sustained into the long-term, with congestion observed in London gradually rebounding five years after scheme introduction to return to pre-policy levels (Metz, 2018). Outside of this expressed policy aim, researchers also contemplated whether congestion charges produced ancillary consequences. With the substantial impact of poor air quality on public health becoming widely known, research evaluating whether congestion charges can also lower concentration levels of local air pollutants has increased in prominence. The findings of these studies are somewhat mixed, with applications showing partial evidence of a reduction in concentrations (Atkinson et al., 2009) whilst others identify increased levels (Qin et al., 2023) plausibly due to the increased use of diesel vehicles (Green et al., 2020). To summarise, impact analysis of congestion charging has tended to focus on immediate implications such as reductions in car use and the resulting effects this has on ancillary issues such as pollution levels. Limited work has so far considered whether congestion charging also impacts car ownership, with only one study conducted on Madrid's private vehicle restrictions by Gonzalez et al. (2021) finding a negative impact.

2.2. Car ownership

Given their centrality in the operation of most developed economy transport systems, determining what factors affect car ownership and how such ownership can be controlled is a topic that has received considerable academic attention. The topic has been considered from multiple disciplinary perspectives, with prominent applications of economics, geography, and sociology (Anowar et al., 2014; de Jong et al., 2004). The determinants of car ownership can be loosely grouped around the individual characteristics of owners, the wider environmental circumstances in which the vehicles are located, and the presence of government policies which may affect ownership decisions.

In terms of individual characteristics, the income level of a household is a primary factor, with car ownership typically expanding as incomes rise (Dargay, 2002). The composition of a household also plays a role, with the presence of children often corresponding to increased rates of car ownership (Nolan, 2010). This taps into a wider issue surrounding life events that can motivate alterations in car ownership, with employment changes and retirement representing other such triggers (Clark et al., 2016; Karjalainen et al., 2023). With commuting being one of the primary drivers of miles-travelled, there is a tendency with individuals that have relatively long commutes to have a greater likelihood of being car drivers (Potoglou and Kanaroglou, 2008; Habib, 2014), though negative relationships between these two variables have also been observed (Shen et al., 2016). In recent years, there has been a reduction in driving licence attainment amongst young adults in some developed economies (Delbosc, 2017). This creates an interesting generational effect, whereby millennials exhibit lower car ownership levels compared to their parents. Several interrelated factors provide plausible explanations for this generational difference, with delayed adulthood, increased costs of motoring for younger drivers, and a growing emphasis of digital connectivity over physical proximity likely playing a role.

The environmental circumstances in the vicinity of households also affect car ownership (Sabouri et al., 2021). The urban-rural spectrum provides a coarse explanation, with car ownership being higher in out-of-town locations and reducing as you move towards dense city environments. This gradient is motivated by a series of spatial and organisational issues. The provision of public transport, such as bus routes and light-rail networks, tends to reduce car ownership levels as

residents have viable transit alternatives (Zhang et al., 2017; Holmgren, 2020; Mulalic and Rouwendal, 2020). The supply of parking opportunities available to a household also strongly affects car ownership, with the presence of garages, driveways, and on-street parking provisions having a positive influence (Guo, 2013).

Of particular relevance to the purpose of this study is how car ownership can be controlled by the government. The application of fiscal policies, which alter the cost implications of owning and operating cars, have been evaluated extensively in the past. At a national level, this can involve alterations to fuel duties (for car operation) alongside changes to registration and circulation taxes (for car ownership). In terms of fuel duties, studies that compare countries in terms of their car ownership levels find no significant impact of fuel price (Chamon et al., 2008). The situation for registration and circulation taxes is somewhat different, with the changes made in Ireland to these fiscal measures considerably altering the composition of new cars being registered (Hennessy and Tol, 2011). At a local level, the application of regulations to park cars on public streets has been examined. The introduction of paid parking for visitors in Barcelona has been found to cause an increase in car ownership amongst residents by 2.9% (Albalade and Gragera, 2020) while increasing the waiting list time to acquire parking permits by one year in Amsterdam reduced car ownership rates by 2% (De Groot et al. 2016).

Identifying government policies that can discourage car ownership, especially in urban environments, is a topic that is rising up the social agenda (Chevallier et al., 2023). Here, the impact of new transport options in urban settings over car ownership has recently received significant attention. Car clubs and ride-sourcing applications provide flexible access to cars and can release households from the financial burdens of car ownership. Studies examining the introduction of car clubs in both Germany and the United Kingdom find that they cause reductions in car ownership post implementation (Kolleck, 2021; Bilgin et al., 2023). Similarly, installing bicycle sharing schemes has been found to reduce car ownership amongst nearby households in the United States (Basu and Ferreira, 2021) and introducing ride-sourcing to a city has been found to reduce car ownership rates in India (Wadud and Namala, 2022). While local policies have been found effective at discouraging car ownership in particular areas, they can generate unintended spillover effects. The work of Zheng et al. (2021) sheds light on this issue, which found that the implementation of a car license plate lottery in Beijing led

to increased car registrations in nearby cities. It is this subset of government policy intended to reduce car ownership that the research reported in this paper contributes the most to, figuring out how the supremacy of the car in urban environments can be deemphasised.

3. Methods

3.1. London Congestion Charge

Introduced in 2003, the London Congestion Charge is a closed boundary regulation, which charges drivers a set daily fee if they enter the zone during the hours of operation. The daily fee was initially set at £5, though it has iteratively increased to £15 in 2023. The charge is in operation during 07:00 to 18:00 on weekdays and 12:00 to 18:00 on weekends. The boundary of the congestion charge is displayed in Fig. 1, with the zone covering 21.4 km² and containing a total of 31,939 private car registrations in 2001. Over the years, the congestion charge zone has undergone a series of modifications. The most prominent modification was the Western Extension, which expanded the zone over the borough of Kensington and Chelsea. This Western Extension was implemented in 2007 and subsequently removed in 2011.

Since its introduction, the London Congestion Charge has received considerable attention in terms of academic research. Aligned closely to the study reported in this paper, quasi-natural experiments have been applied to identify the impacts the policy has induced. Examples of such work include the consequences of the policy on traffic volumes within the charging zone, with a 4.4% reduction being identified (Ouali et al., 2021). The reduction of traffic volumes creates a more pleasant living environment for residents and makes the area more desirable. This is visible through residential property values, with Tang (2021) detecting a circa 3% uplift post implementation. Similarly, reducing traffic volumes can create benefits for traveller safety (Ding et al., 2021), with Green et al. (2016) showing that the introduction of the congestion charge in London resulted in a 22% reduction in the rate of traffic collisions in the charging zone.

3.2. Data

The vehicle licensing statistics database curated by the Department for Transport (2023) in the United Kingdom is the primary data source

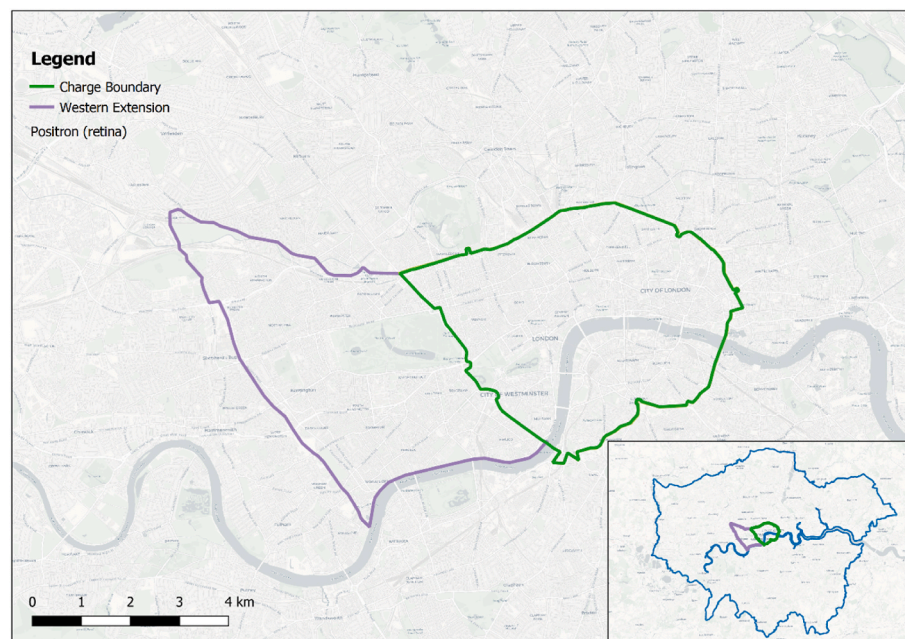


Fig. 1. Map of the London Congestion Charge covering the boundary of the resident discount (in green) alongside the defunct Western Extension (in purple). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

for this project. This database holds a record of each vehicle registered for use on the road, representing a complete account of the total population of cars. The number of privately owned cars registered to residents represents the outcome variable in the analysis. Note that the outcome variable does not consider fleet registrations (i.e., cars owned by firms). The omission of fleet registrations is to avoid complications of corporate cars registered to a particular location (e.g., a firm's headquarters) but used in other locations. Annual registrations from the period 2000 to 2006 are extracted from the database to cover before and after the introduction of the London Congestion Charge. To provide a wider content to the analysis, demographic data from the 2001 census of the population is sourced, covering characteristics of employment activity, travel to work patterns, and population densities (Office of National Statistics, 2001). The census is completed by each household in the United Kingdom and represents a complete account of the population.

All data utilised in the analysis are aggregated to the Output Area level of spatial resolution. Output Areas represent the lowest census geography level in the United Kingdom and typically contain 150 dwellings. Within the Greater London Area, there are 24,140 Output Areas present, whilst the London Congestion Charge when initially introduced covered 575 Output Areas.

3.3. Quasi-natural experimental design

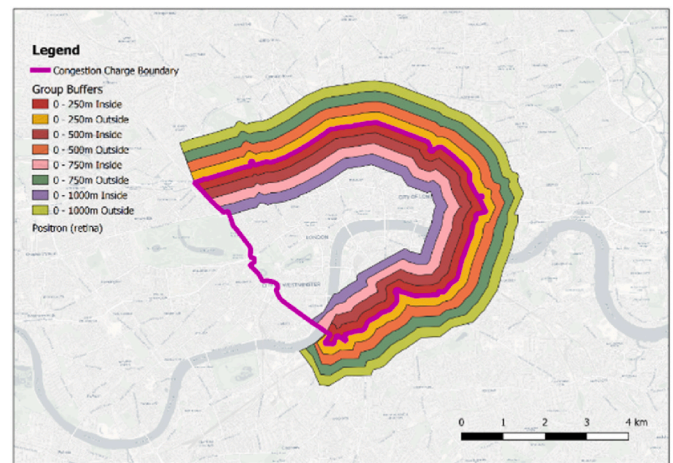
The London Congestion Charge contains a useful feature that facilitates the application of a quasi-natural experiment. Areas located within the charging zone were provided with a 90% resident discount for a single car (i.e., their residents only had to pay a 50p daily charge to cross the zone, as opposed to a £5 fee when the policy was implemented). No such discount existed for areas on the outside of the charging zone, whose residents were required to pay the full fee for entrance. This feature creates treatment and control groups for the policy, with areas inside the charging zone representing the control group whilst those outside represent the treated group.

As mentioned previously, the outcome variable for the analysis is private car ownership. The intuition is that in treated areas, the operating costs of driving a car have increased. After the introduction of the congestion charge, residents in treated areas now need to pay a daily fee to drive into the city centre. If residents in treated areas do not wish to pay this daily fee, then they must avoid the charge through such means as detouring their drive around the charge boundary, delaying their drive to hours outside of the congestion charge operation, changing their mode of travel, or forgoing travel. Indeed, it is residents in treated areas just outside the boundary that arguably have their driving patterns most affected by the introduction of the congestion charge. With operating costs being a significant consideration when drivers consider car ownership (de Jong, 1990; de Jong et al., 2009), it is plausible that the application of the congestion charge would encourage some drivers to select alternative modes of transport to service their mobility needs. This response is most likely for drivers where car ownership is marginal,

where the benefits of owning a car are only slightly better than other options. On the other hand, residents in the control group areas are shielded from the vast majority of the applicable fee for driving across the zone whilst benefitting from reduced traffic volumes within the zone. Such a response has been suggested in past work (Ubbels et al., 2008, as cited in Li and Hensher, 2012) whereby 2% of survey respondents stated they would sell their car due to the imposition of a road user charge. Given this situation, the hypothesis for the research is that the introduction of the congestion charge will negatively impact private car ownership levels in the treatment group areas.

One important consideration in the experimental design is how quickly the impact of the policy will take. Given that public conversation surrounding the congestion charge began in the years leading up to the introduction of the policy (as outlined in the policy timeline of Fig. 2, reported in Leape (2005)), it is plausible that residents in the treatment group areas may have decided to reduce their car ownership before the date of implementation (i.e., an anticipatory effect). Conversely, it is plausible that residents in the treatment group areas may have held off on reducing their car ownership until a certain trigger event occurs (i.e., a delayed effect). Such situations could be residents deciding not to renew a lease on a car when it expires after the introduction of the congestion charge or not replacing a car that has become uneconomical to repair. To provide some leeway in the analysis, a 1-year buffer is utilised around the introduction of the congestion charge. This buffer means that private car registrations in 2001 and 2005 are compared in the analysis as the before and after periods.

The boundary of the London Congestion Charge is used to delineate the treatment from the control groups within the analysis. Output Areas on the inside of the boundary are assigned to the control group, whilst



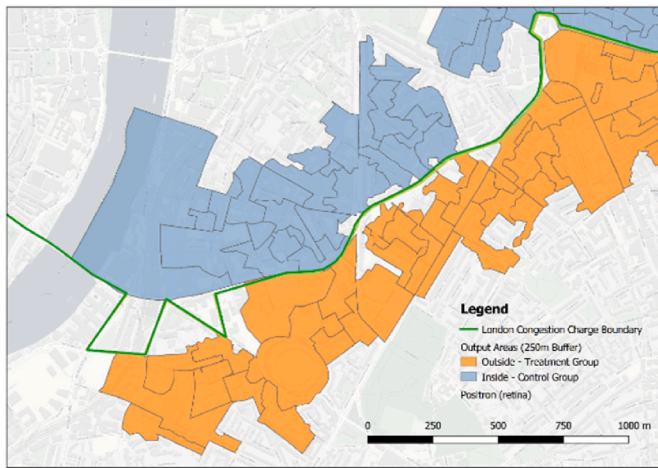


Fig. 4. Map demonstrating the assignment of Outputs Areas to treatment and control groups using the 250m buffer band.

those on the outside are assigned to the treatment group. Distance buffers from the boundary are used in the group assignment, increasing from 250 m to 1000 m in 250-m increments. This allows the analysis to consider the geographic reach of the policy's impact by specifying four separate models using the different buffer distances. Output Areas that intersect these distance buffers are assigned to the relevant group. Output Areas positioned on the westward side of the boundary are excluded from the analysis. This exclusion is due to the Western Extension of the congestion charge that was introduced between 2007 and 2011 (depicted in Fig. 4), which could obscure the policy's signal. In circumstances where an Output Area intersects the charging zone boundary, the Output Area is discarded from the analysis. Figs. 3 and 4 provide an overview of these buffer bands and Output Area assignment, respectively.

3.4. Analytical procedure

To apply the quasi-natural experiment, a difference-in-differences (DiD) procedure is followed. This procedure suits the time series nature of the study and provides an estimate of the causal impact of the policy in terms of the average treatment effect on the treated.

One of the particular strengths of a DiD application is that it does not require randomisation in the experiment. This strength depends on a fundamental assumption: the treatment and control groups exhibit parallel trends in the outcome variable before the policy is implemented. Achieving the parallel trends assumption means that no time-variant confounders (i.e., variables not observed in the experimental design) are impacting the outcome variable across the treatment and control groups (e.g., pulling them in different directions). To evaluate the parallel trends assumption, two checks are implemented.

The first check is a visible inspection of the mean car ownership rates in the treatment and control groups observed before the introduction of the policy. The second check is the application of an event study through a two-way fixed effects regression model. In this check, a panel analysis is conducted over the entire time period for which data is available (i.e., 2000–2006) with the inclusion of dummy variables that interact time-to-policy with treatment group. Observations in the year 2003 are used as the reference category. To support the parallel trends assumption, the expectation is that all the dummy variables interacting time-to-policy with treatment are insignificant until after the policy has been implemented.

$$y_{it} = \alpha + \beta_1 (Time_t^{-3} \times Treatment_i) + \beta_2 (Time_t^{-2} \times Treatment_i) + \dots + \beta_6 (Time_t^{+3} \times Treatment_i) + u_i + \delta_t + \varepsilon_{it} \quad (1)$$

This event study is summarised in Eq. (1), where,

y_{it} represents observations of private car ownership in the i th Output Area in the t th year

α represents the model constant

$(Time_t^{-3} \times Treatment_i)$ represents an interactive dummy variable which takes the form of 1 for observations of treated Output Areas in year 2000 (i.e. 3 years before policy implementation)

u_i represents the unit specific effect

δ_t represents the time specific effect

ε_{it} represents the model residual

For the DiD analysis, a baseline random effect model is initially specified. This follows a two-by-two setup, whereby the panel has two time periods (i.e., before and after, 2001 and 2005) and two groups of observations (i.e., control and treatment). This baseline random effect DiD is summarised in Eq. (2), whereby the components in addition to those which featured in Eq. (1) are $Treat \times Time_{it}$, which represents a dummy variable denoting the interaction between treatment and time, β_1 , which represents the causal impact of the congestion charge over private car ownership, $Treat_i$, which is a dummy variable denoting the experimental group assignment of the i th Output Area, $Time_{it}$, which is a dummy variable denoting the time period (i.e. before or after) of the i th Output Area in the t th year, and $Demo_i$ which represents observations of time-invariant demographic variables.

$$y_{it} = \alpha + \beta_1 (Treat \times Time)_{it} + \beta_2 Treat_i + \beta_3 Time_{it} + \beta_4 Demo_i + u_i + \varepsilon_{it} \quad (2)$$

This baseline DiD is a-spatial in nature and does not account for the geographical aspect of the data used in the analysis. Using spatial data in panel regression models can introduce bias through the occurrence of spatial autocorrelation, where observations that are geographically nearby tend to display similar values (e.g. of private car ownership) due to the co-occurrence of supporting conditions (e.g. land-use and population demographics). If not corrected, this bias can lead to estimation of misleading model parameters. To alleviate this potential bias, the baseline a-spatial DiD is extended through additional geographic components. A multitude of spatial panel regression model variants are available to specify the DiD procedure. To guide the selection of the most suitable variant, a series of diagnostics are applied, which are summarised in Table 1. The first diagnostic evaluates whether spatial autocorrelation is present in the outcome variable: the number of private cars registered per Output Area. The Moran's-I test calculates the correlation between this outcome variable and its spatial lag, determining whether values of private car ownership in any given Output Area are significantly related to those values observed in neighbouring Output Areas (Bivand and Wong, 2018). The spatial lag of the private car ownership is calculated through a spatial weights matrix that follows a queen contiguity process, where spatial units are assigned as neighbours if they share a line or a point border. For all buffer bands, the test indicates that spatial autocorrelation is present which could bias the analysis if left uncontrolled. This test result supports the decision to extend the DiD analysis through the inclusion of spatial effects. Such extensions are typically referred to as spatial difference-in-differences models (i.e., SpDiDs) (Debe et al., 2014; Delgado and Florax, 2015).

Spatial effects can be incorporated into panel regression models in two primary ways. The first is through including a spatial lag of the

Table 1

Diagnostics covering tests of spatial dependence, panel effects, and spatial model extensions.

Buffer	Moran's-I	Spatial Hausman	Robust LM (lag)	Robust LM (error)
250 m	0.216 ^a	2.021	9.176 ^a	0.878
500 m	0.307 ^a	3.239	8.219 ^a	3.001
750 m	0.278 ^a	2.540	22.196 ^a	0.049
1000 m	0.292 ^a	0.665	19.795 ^a	0.670

^a p-value < 0.01.

model's dependent variable as a regressor (typically referred to as the spatial autoregressive effect) whilst the second is through incorporating a spatial lag of the model's residual in the error term (typically referred to as the spatial error effect) (Elhorst, 2014). The Robust Lagrange Multiplier tests provide guidance on which spatial effect to select (Anselin et al., 1996). For all buffer bands, the test returns significant results for the spatial lag extension (i.e. Robust LM (lag)) while the equivalent results for the error extension are insignificant. This implies that specifying a model which incorporates a spatial autoregressive effect is preferable.

The final model diagnostic covers which panel model variant to select. Spatial extensions of the conventional Hausman test are applied to determine whether a fixed or random effects panel model would be most appropriate (Mutl and Pfaermayr, 2011). This Hausman test compares the coefficients estimated for both fixed and random effects model variants to determine if they are significantly different from one another. If the null hypothesis is confirmed, then the random effects variant provides consistent and efficient estimates of the model coefficients. If the alternative hypothesis is confirmed, then only the fixed effects model provides consistent coefficient estimates. For all the buffer bands, the spatial Hausman test returns an insignificant result, confirming the null hypothesis. This result supports the selection of a random effects panel model. The final DiD model specified in the analysis is summarised in Eq. (3), whereby the components in addition to those which featured in Eqs. (1) and (2) are Λ , which represents the spatial autoregressive term, and Wy_{it} , which represents spatially lagged observations of private car ownership in the i 'th Output Area in the t 'th year.

$$y_{it} = \alpha + \Lambda Wy_{it} + \beta_1 Treat_i + \beta_2 Time_{it} + \beta_3 (Treat \times Time)_{it} + \beta_4 Demo_i + u_i + \varepsilon_{it} \quad (3)$$

To summarise, the final model variant selected to specify the DiD procedure is a random effects panel with spatial autoregressive and unit specific effects. This model variant is typically referred to as the Spatial Autoregressive Random Effects model (i.e., a SARRE model). Models of this variety are starting to appear in transport studies as researchers become aware of their strengths in identifying causal impacts, with applications present for mass transit's impact over property values in both Denver (Bardaka et al., 2018) and Singapore (Diao et al., 2017). A binary queen continuity spatial weights matrix is utilised to calculate the spatial autoregressive term, which assigns spatial units as neighbours if they share either a line or a point border. The plm and splm packages in the R programming environment are employed to estimate the models (Croissant and Millo, 2008; Millo and Piras, 2012). A log-normal format is followed, with observations of the outcome variable transformed into its natural logarithm to ensure positivity.

3.5. Sensitivity testing

The analytical procedure described involves a considerable number of empirical choices that can have bearing over the outcomes produced. In part, the selection of appropriate options is guided by the application of diagnostics to determine which deliver a robust study design. To ascertain the stability of the analysis in terms of important empirical choices, three different sensitivity exercises are conducted.

The first sensitivity test is incorporated into the main method and involves the use of different buffers which spread out from the outer boundary of the congestion charge to assign Output Areas to treatment and control groups. This allows the analysis to consider the geographic reach of the policy and detect whether the impact of the policy diminishes over a short distance (i.e., out to 1 km).

The second sensitivity test relates to the selection of suitable spatial weights to calculate the spatial interaction terms. In general, these can come in three different forms and be based on either contiguity, distance, or k-nearest neighbours. In the sensitivity analysis, both distance-

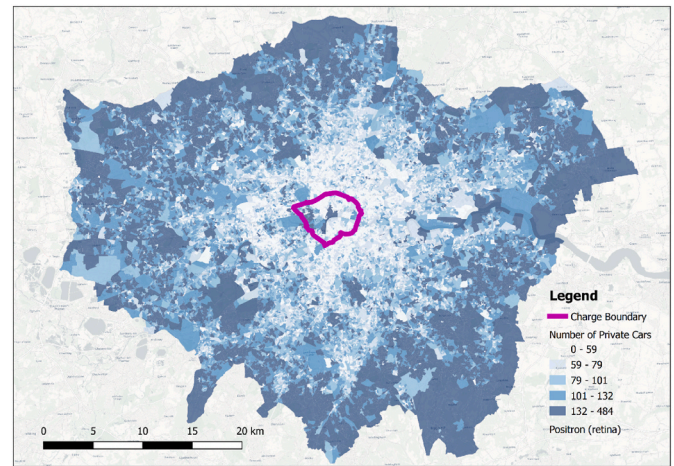


Fig. 5. Map of the number of private cars registered in each Output Area of Greater London in 2003.

based and k-nearest spatial weights are substituted into the analysis and compared to the default contiguity approach to assess whether this choice substantially impacts the outcomes of the experiment.

The third sensitivity test covers the selection of the before and after periods used in the analysis. In the main analysis, a 1-year buffer is used to allow for the occurrence of anticipation and lag affects in the stimulus-response mechanism. To test the consequence of this decision, the analysis is repeated using no time buffers (i.e., using 2002 as the before period and 2004 as the after period). This sensitivity test evaluates whether the findings of the analysis are resistant to alterations in the time period selected and if anticipatory or lag effects are necessary for impacts to be identified.

4. Results and discussion

4.1. Descriptive analysis

Fig. 5 displays the geographic variation in private car ownership across Greater London in 2001, with deeper shades signifying higher ownership rates. A spatial gradient is visible, whereby Output Areas closer to the congestion charge typically have fewer cars registered as compared to Output Areas located at the city's periphery. This corresponds to past work that has examined the socio-environmental determinants of car ownership, with suburban areas on the outskirts of

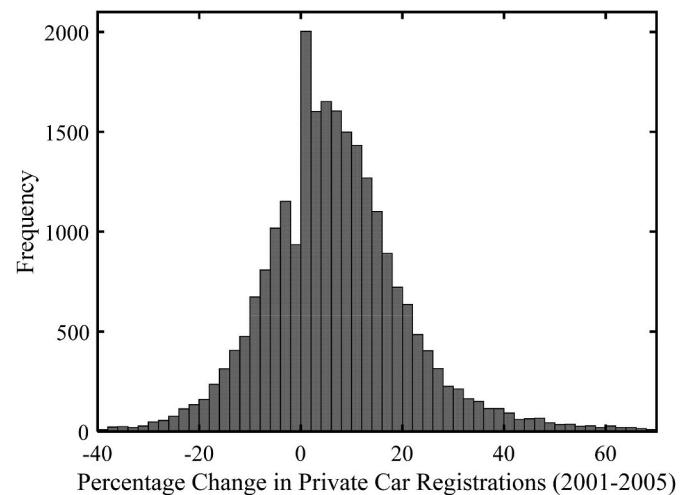


Fig. 6. Histogram of the change in the number of private cars registered in each Output Area of Greater London between 2001 and 2005.

Table 2

Percentage change in private car ownership rates between 2001 and 2005 for Output Areas assigned to the treatment and control groups by buffer range.

Buffer Band	Experimental Group	Percentage Change
250 m	Treatment (outside)	1.80%
	Control (inside)	11.51%
500 m	Treatment (outside)	2.91%
	Control (inside)	11.89%
750 m	Treatment (outside)	2.96%
	Control (inside)	14.21%
1000 m	Treatment (outside)	2.77%
	Control (inside)	15.62%

cities having the conditions more conducive to cars such as dedicated parking provision, lower provisions of public transport, and households with longer commutes (Sabouri et al., 2021; Zhang et al., 2017; Guo, 2013).

Fig. 6 charts the change in private car ownership levels before and after the introduction of the London Congestion Charge across all the Output Areas of Greater London. During this time period, the majority of Output Areas experienced expansions in their private car fleets. However, a sizable minority of Output Areas did record reductions in ownership in the range of 0 to –20%. This tendency for car fleet expansion during this time period coincides with the economic conditions that were present in the United Kingdom at the time, with household incomes growing up until the global financial recession.

Table 2 notes the average percentage change in private car ownership in the control and treatment groups between 2001 and 2005 by buffer band. Across all buffer bands, the number of registered cars in the treatment group remained reasonably stable whilst in the control group car ownership increased. The degree of ownership expansion in the control and treatment groups is similar across the different buffer bands, implying that the spatial reach of this policy does not diminish over short distances.

4.2. Parallel trends checks

To evaluate the parallel trends assumption which is crucial in the identification of causal impacts in difference-in-differences, two checks are applied. The first check is a visual inspection of the outcome variable trends before policy implementation. Fig. 7 charts this time series for the 250 m buffer, with highly similar trends apparent for the charts of the

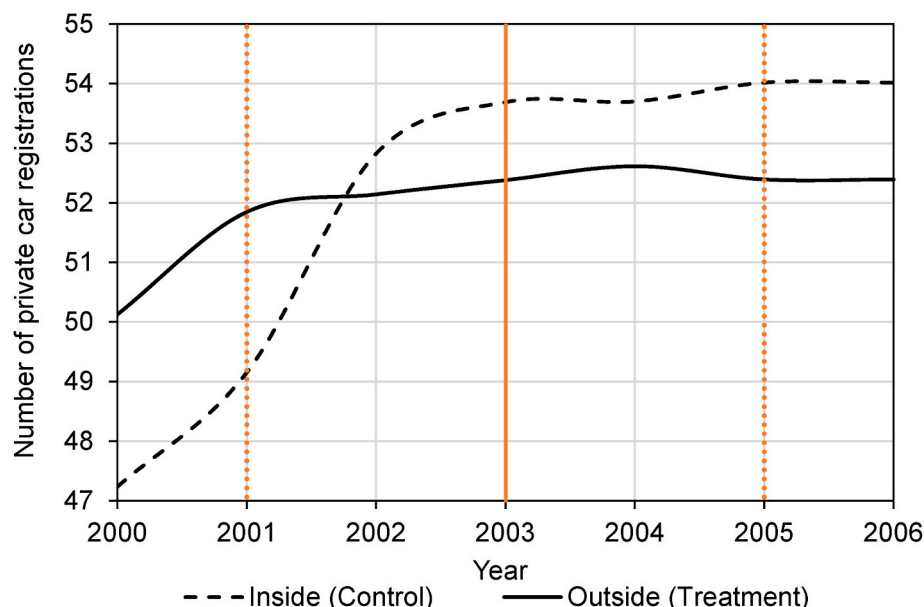


Fig. 7. Timeseries of average number of private car registrations in Output Areas located in the treatment and control groups defined by a 250m buffer.

Table 3

Event Study Analysis using interaction terms between time to treatment and treatment group.

	Buffer Band (meters)			
	250	500	750	100
2000 × Treatment	–0.030 (0.020)	–0.016 (0.014)	–0.005 (0.013)	–0.002 (0.011)
2001 × Treatment	–0.000 (0.017)	–0.001 (0.012)	0.009 (0.011)	0.012 (0.010)
2002 × Treatment	–0.008 (0.014)	–0.005 (0.010)	0.003 (0.009)	0.005 (0.008)
2004 × Treatment	–0.010 (0.011)	–0.001 (0.009)	–0.009 (0.008)	–0.012 (0.007)
2005 × Treatment	–0.082 ^a (0.017)	–0.074 ^a (0.013)	–0.078 ^a (0.011)	–0.083 ^a (0.011)
2006 × Treatment	–0.095 ^a (0.024)	–0.066 ^a (0.019)	–0.073 ^a (0.016)	–0.072 ^a (0.015)

^a p -value < 0.01.

other buffer categories. One issue to point out is that, between 2000 and 2001, the treatment and control groups had upward trending car ownership rates with similar slopes.

The second check involves the application of an event study analysis, with the results presented in Table 3. Four separate event studies are reported covering the different buffer bands to evaluate the stability of the results to changes in distance from the congestion charge outer boundary. For each of the event studies, the results are highly consistent. None of the interaction terms between time-to-event and treatment group for those years leading up to the policy are significant (i.e., coefficients on the variables 2000 × Treatment through 2002 × Treatment). This result indicates that the control and treatment groups did not exhibit significantly different trends in the years preceding policy introduction.

The results of these checks do not refute the parallel trend assumption, indicating that no time-invariant confounders are impacting private car ownership of the treatment and control groups in different ways before the policy is implemented. A visual inspection of Fig. 7 shows that, at the start of the time-series, private car ownership in the control group was beneath that of the treatment group, while this situation had reversed by the end of the time series. Private car ownership in the control group overtakes that of the treatment group in the year leading up to the introduction of the congestion charge. This may indicate an anticipatory response in the treatment group, which pre-emptively

Table 4

Baseline random effect difference-in-difference (DiD) models estimating the impact of congestion charging on private car ownership.

Parameter	Buffer band (m)			
	250	500	750	1000
Constant	3.659 ^a (-0.185)	3.709 ^a (-0.142)	3.810 ^a (-0.125)	3.867 ^a (-0.112)
DiD Variables				
Treatment Dummy	0.004 (-0.041)	0.021 (-0.032)	0.043 (-0.028)	0.044* (-0.026)
Time Dummy	0.080 ^a (-0.016)	0.083 ^a (-0.013)	0.100 ^a (-0.011)	0.105 ^a (-0.011)
Treatment x Time Dummy	-0.082 ^a (-0.023)	-0.073 ^a (-0.017)	-0.088 ^a (-0.015)	-0.096 ^a (-0.014)
Time Invariant Variables				
Economically Active (%)	0.009 ^a (-0.002)	0.006 ^a (-0.001)	0.005 ^a (-0.001)	0.004 ^a (-0.001)
Drive to Work (%)	0.022 ^a (-0.007)	0.030 ^a (-0.005)	0.026 ^a (-0.004)	0.027 ^a (-0.004)
Flats (%)	-0.007 ^a (-0.001)	-0.006 ^a (-0.001)	-0.006 ^a (-0.001)	-0.006 ^a (-0.001)
Population Density	0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 ^a (-0.0001)	-0.0003 ^a (-0.0001)
Unit Specific Random Effect Term				
<i>u</i>	0.118	0.127	0.124	0.129
Model Statistics				
Observations	718	1262	1738	2224
R ²	0.141	0.132	0.131	0.133
Adjusted R ²	0.132	0.127	0.127	0.13
F Statistic	116.227 ^a	190.572 ^a	259.751 ^a	339.096 ^a

^a *p*-value <0.01.

restricts private car ownership levels in 2002 in expectation of the introduction of the congestion charge at the start of 2003 (policy implementation was February 17th).

4.3. Difference-in-differences analysis

The results of the baseline random effects difference-in-differences (i.e., DiD) models are summarised in Table 4, with the columns reporting the estimated variable coefficients and standard errors (in parentheses) across the different buffer bands. The baseline model includes both unit specific fixed effects, with the causal impact of the policy estimated by the Treatment × Time Dummy. For each of the different buffer bands, the causal impact of the policy is significant and negative.

The extension of the baseline analysis through the use of spatial difference-in-differences (i.e., SpDiD) is reported in Table 5. This takes the form of a spatial autoregressive with unit specific random effects model (i.e. a SARRE model). The coefficients calculated for the time-invariant variables denoting the demographic features of the Output Areas follow expectations. The variables recording the proportion of residents that are economically active (i.e., either in employment or actively seeking employment) and drive and car to work have positive coefficients, implying that as the workforce enlarges and people select cars for their commute, the number of private cars owned tends to expand. The proportion of houses classified as flats holds a negative coefficient in all models. With flats generally having a lower provision of dedicated car parking, they can be more attractive to carless households. The variable measuring the population density of the Output Area is significant in the model which uses 1000 m buffers but not so in the models with smaller buffers, providing partial support to the view that higher population densities are associated with lower private car ownership.

The coefficient for the spatial autoregressive term (i.e., λ , the spatially lagged dependent variable) is significant in all models, indicating that spatial dependence is present within the analysis. In the model using the 1000 m buffer band, the dummy variable for Treatment has a positive coefficient. This result indicates that being assigned to the treatment group is associated with a higher rate of car ownership,

Table 5

Spatial difference-in-difference (SpDiD) models estimating the impact of congestion charging on private car ownership.

Parameters	Buffer band (meters)			
	250	500	750	1000
Constant	3.114 ^a (0.008)	2.890 ^a (0.135)	2.879 ^a (0.118)	2.960 ^a (0.107)
DiD Variables				
Treatment Dummy	-0.008 (0.040)	-0.001 (0.031)	0.016 (0.027)	0.015 ^a (0.025)
Time Dummy	0.068 ^a (0.016)	0.062 ^a (0.013)	0.073 ^a (0.011)	0.077 ^a (0.011)
Treatment × Time Dummy	-0.072 ^a (0.023)	-0.057 ^a (0.017)	-0.066 ^a (0.015)	-0.072 ^a (0.014)
Time Invariant Variables				
Economically Active (%)	0.008 ^a (0.002)	0.005 ^a (0.001)	0.004 ^a (0.001)	0.003 ^a (0.001)
Drive to Work (%)	0.017 ^a (0.007)	0.026 ^a (0.005)	0.022 ^a (0.004)	0.023 ^a (0.004)
Flats (%)	-0.007 ^a (0.001)	-0.006 ^a (0.001)	-0.006 ^a (0.001)	-0.005 ^a (0.001)
Population Density	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.0002 ^a (0.0001)
Spatial Autoregressive Term				
λ	0.175 ^a (0.046)	0.245 ^a (0.036)	0.269 ^a (0.031)	0.263 ^a (0.028)
Unit Specific Random Effect Term				
<i>u</i>	4.679 ^a (0.550)	4.956 ^a (0.443)	4.732 ^a (0.360)	4.746 ^a (0.317)
Model Statistics				
Observations	716	1260	1736	2222
Log-Likelihood	-92.060	-169.016	-232.931	-363.229

^a *p*-value <0.01.

though this effect is not found at smaller buffer distances. Given the spatial patterns of private car ownership illustrated in Fig. 4, this result seems in keeping with expectations as locations further removed from the urban core typically have higher rates of ownership. For all buffer bands, the dummy variable Time proves to be significant. This result implies that there is a general tendency for private car ownership to increase through the period of the analysis.

The causal impact of the introduction of congestion charging over private car ownership is measured through the Treatment × Time dummy variable. Here, a stable result is present across both a-spatial and spatial DiDs and different buffer bands. The results of the analysis indicate that the policy causes a reduction in private car ownership. The magnitude of this reduction in the spatial difference-in-differences models ranges from a low of -5.54% in the model using 500 m buffers to a high of -6.95% in the models using 250 m and 1000 m

Table 6

Sensitivity test using different spatial weights matrices as part of the SpDiD using the 250m buffer to evaluate model stability.

Parameters	250 m		
	Contiguity	K-Nearest	Distance
DiD Variables			
Treatment Dummy	-0.008 (0.040)	-0.003 (0.039)	-0.003 (0.039)
Time Dummy	0.068 ^a (0.016)	0.070 ^a (0.016)	0.059 ^a (0.016)
Treatment × Time Dummy	-0.072 ^a (0.023)	-0.073 ^a (0.023)	-0.077 ^a (0.023)
Model Components			
Demographic Characteristics	✓	✓	✓
Spatial Interaction Effect	✓	✓	✓
Unit Specific Random Effect	✓	✓	✓
Model Statistics			
Observations	718	718	718
Log-Likelihood	-92.060	-101.823	-92.060

^a *p*-value <0.01.

buffers. The estimates of the causal impact derived from the a-spatial DiDs are always higher than their counterparts in the spatial DiDs. For example, the causal impact estimated in the a-spatial DiD using 1,00m buffer is −9.15%, as compared to −6.95% using the spatial DiD.

4.4. Sensitivity analysis

The three sensitivity tests applied in this study showcase how the outcomes of the analysis respond to changes in key analytical choices. The results of the first test, which uses different buffer distances to assign Output Areas to treatment and control groups, are visible in [Tables 4 and 5](#). Here, the results of the sensitivity test are reasonably stable. The variable coefficients in the models using different buffer bands are similar in terms of magnitude, with the only substantive difference being that the treatment dummy variable in the SpDiD is significant in the model using 1000 m buffers while being insignificant in the remaining models.

The results of the second sensitivity test are reported in [Table 6](#) and cover different approaches to calculating the spatial lags used in the SpDiDs. A contiguity spatial weights matrix (i.e., neighbours are assigned by shared borders) is used in the main analysis, with *k*-nearest and distance-based alternatives evaluated. The results suggest that the different approaches to calculating the spatial weights have little bearing over the outcomes of the analysis. The independent variables included in the model generally share the same degree of significance and similar coefficient sizes across the different model variants. The only substantive difference that is detected is for the spatial autoregressive term, where the estimated coefficient is substantially larger in the model using distance based spatial weights. This is likely motivated by the distance based spatial weights typically assigning considerably more spatial units as neighbours compared to the alternatives.

The final sensitivity test examines the choice of before and after periods to base the analysis on. In the main analysis, a 1-year buffer is used to allow for the occurrence of anticipation and lags in the stimulus-response relationship. Here, the check removes this buffer and uses 2002 as the before period and 2004 as the after period (i.e., the years immediately surrounding the implementation of the congestion charge in 2003). Using this new timeframe, two-way random effect spatial DiD models are specified with the results summarised in [Table 7](#). For the models using all buffers distances for the assignment of experimental groups, no casual effect of the policy on private car ownership is identified (i.e., the estimated coefficient on the Time × Treatment variable is insignificant). This indicates that the analysis is sensitive to the selection of before and after periods and that using a narrower window of time for the response to the policy to take affect leads to no impact being visible.

Table 7
Sensitivity test using 2002 and 2004 as the before and after period in the spDiD model.

Parameter	Buffer band (meters)			
	250	500	750	1000
DiD Variables				
Treatment Dummy	−0.063 (0.038)	−0.055 (0.030)	−0.036 (0.026)	−0.038 (0.024)
Time Dummy	0.008 (0.011)	0.009 (0.009)	0.018 ^a (0.008)	0.017 ^a (0.007)
Treatment × Time Dummy	−0.006 (0.017)	0.002 (0.013)	−0.010 (0.011)	−0.013 (0.010)
Model Components				
Demographic Characteristics	✓	✓	✓	✓
Spatial Interaction Effects	✓	✓	✓	✓
Unit Specific Random Effects	✓	✓	✓	✓
Model Statistics				
Observations	718	1262	1738	2224
Log-Likelihood	13.360	−4.299	−17.725	−40.505

**p-value <0.01.

^a p-value <0.05.

Given that changes in car ownership are considered as a longer-term response to the introduction of congestion charge, the use of the 1-year buffer seems well justified in the main analysis. The visual inspection of the time series in private car ownership between the two experimental groups presented in [Fig. 7](#) does suggest that an anticipatory effect is present, whereby private car ownership rates in the treatment group are dampened in 2002 leading up to the introduction of the congestion charge in early 2003.

5. Conclusions

The purpose of this study is to determine whether the application of a congestion charge reduced private car ownership rates. To achieve this aim, robust statistical modelling is applied to a real-world example using reliable outcome data. The unique features of the London Congestion Charge provide a valuable natural experiment that is utilised to determine the impact that this policy has over car ownership. Applying difference-in-differences techniques in this natural experiment improves the credibility of the impact assessment over more descriptive analysis, giving policymakers confidence in using these results in their evaluations. Extending the difference-in-differences approach by including spatial interaction effects is a further refinement of the technique, showcasing how the method continues to advance.

The analysis outcomes are highly consistent here and indicate that applying a congestion charge reduces private car ownership rates in nearby locations. This impact is detected out to 1 km from the congestion charge's outer boundary, showcasing the considerable reach of the policy. The extent of the causal impact diminishes in the spatial difference-in-differences models as compared to their a-spatial counterparts. This situation is also present in other applications ([Diao et al., 2017](#); [Basu and Ferreira, 2021](#)), whereby the impact of transport policy reduces when spatial variants of difference-in-differences are applied. While a reduction in the impact by 2.21% does not sound like much overall (i.e., −9.15% to −6.95%, comparing the estimates derived using the 1,000m buffer), this represents a percentage change between the two values of −24.1%. If the spatial extension of the difference-in-differences procedure was not applied, then the true causal impact of the policy could have been overestimated by close to a quarter. It is important to note here that this impact is identified from the implementation of a £5 daily congestion charge. If the congestion charge was of a greater value, the impact would likely be greater.

While the London Congestion Charge has been in effect for two decades, it is being periodically revised to respond to changing circumstances. This is visible in the recent introduction of the Ultra Low Emission Zone regulations, which impose additional daily charges on drivers of cars that do not comply with certain emission standards. In addition to modifications of existing schemes, other cities are still contemplating the introduction of congestion charging as the fundamental need to control traffic volumes remains a high priority. The research presented in this paper is instructive to policymakers considering the adaptation of existing congestion charges or the introduction of new schemes. This instruction is by demonstrating the wider ancillary impacts such schemes can provide, which can feed into the development of business cases that weigh up the costs and benefits attributable to scheme operation. These ancillary impacts may not justify the introduction of congestion charging in their own right but could tip the balance in the favour of these progressive forms of transport policy.

At a broader level, this study showcases the type of analysis that can be done through the provision of reliable outcome data at a high level of spatial resolution, which is periodically refreshed. The increasing availability of spatially located data is opening more doors for analysis that looks at the impact of policies and other interventions. Whilst the focus of this work is on private car ownership, other transport-related outcome variables that could be considered are public transport ridership, pedestrian footfall, mode choice, link speeds, and speed limit violations, to name but a few. Scholars who have an interest in such outcome areas may find it

useful to consider whether data is becoming available that makes quasi-natural experiments such as difference-in-differences possible as it will provide greater confidence in the impact estimates.

CRedit authorship contribution statement

Craig Morton: Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization. **Yasir Ali:** Writing – review & editing, Methodology.

Study limitations

There are five limitations to the empirical design that should be kept in mind while interpreting the results of the analysis.

First, the use of spatial data on private car registrations means that inferences to individual behaviour should be conducted cautiously. It is entirely possible that instead of encouraging residents to reduce car ownership, the introduction of a congestion charge prompted car owning residents to leave the immediate area and be replaced by car-free residents. Such an occurrence would represent spatial sorting through residential self-selection rather than behaviour change. Indeed, it could be a combination of these two motivations at play that cause any impact which is identified. Determining the balance of these underlining motivations cannot be determined using the data evaluated in this study, though other applications in transport studies do focus on unpicking this issue (Ding et al., 2024). As a result, it is safer to discuss the results in terms of outcomes for the population in a close proximity to the congestion charge rather than the behaviour of individual residents.

Second, the estimation of the causal impact of the policy is not without plausible drawbacks. In the main analysis, the empirical design selected is a spatial extension of conventional difference-in-differences. This selection was motivated by the spatial nature of the data used and accounts for the potential bias of spatial autocorrelation. However, other extensions to conventional difference-in-differences are possible that bring different benefits. Examples that have already been applied in transport studies are semi-parametric applications using machine learning techniques (Zhou et al., 2024). Such applications allow the analysis to detect non-linear cause-effect relationships and how these relationships evolve over time.

Third, the demographic variables included in the analysis are quite limited and do not include some pertinent issues that are known to influence car ownership. Notable amongst these is the absence of household income levels, which is typically viewed as the most important characteristics in car ownership. This situation is due to data availability, with demographic characteristics restricted to what is available in the population census of 2001.

Fourth, this analysis is focused on a policy that was enacted twenty years ago, with the circumstances of the policy having changed considerably in the intervening time period. These changes cover revisions to the congestion charge's spatial layout, vehicle access requirements, and pricing. The analysis is unable to determine whether any observed impact on car ownership persist into the longer-term, nor whether the same impact would occur in the present given the considerable changes that have occurred in London.

Fifth, this analysis makes use of spatial buffers for the assignment of experimental groups. While on the surface this allows for a more compelling case to be made for the control of exogenous variables in the study, it can mask geographic features that act as discontinuities to social organisation and thus may bias the results. An example of such a geographic feature is water bodies, that naturally inhibit movement. Future research could examine such choices to determine if alternative forms of experimental group assignment can reveal whether such biases are present and how to account for them.

Conflict of interest

The author is not aware of any conflict of interest that is associated

with this paper.

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Data availability

The authors do not have permission to share data.

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